

t-test

This test is used to compare the mean of two group/sample to see if they are significantly different from each other.

Ex:

- ***Group 1:** 30 students who studied with Method A.
- **Group 2:*** 30 students who studied with Method B.

Assumption in t-test

Independence:

The value of one observation should not influence the value of another observation

-Normality:

The data should be approximately Normally distributed

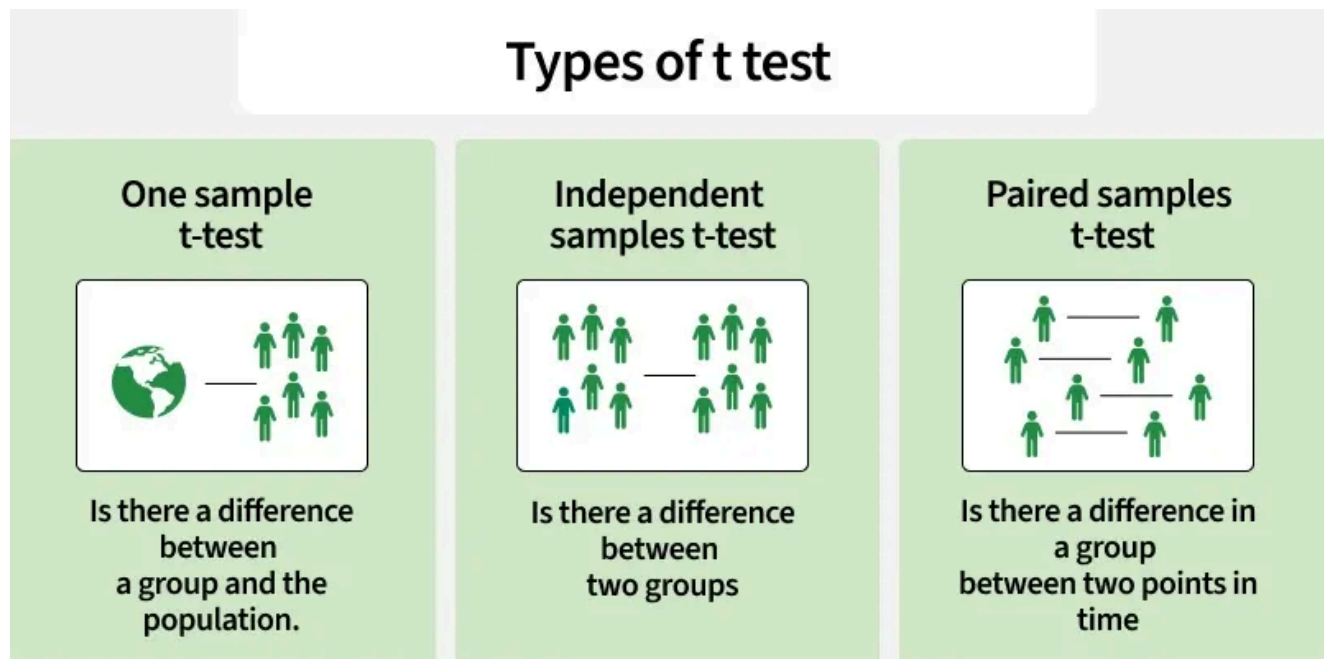
-Homogeneity of Variances:

The variance of two group that are being compared are equal to ensure that the group have a similar spread of values

-Absence of Outliers:

no outlier

Types of t-test



Critical value = significance level / 2

-if the z-test > Critical Value

we reject the null hypothesis

-if the z-test < Critical Value

we do not have enough evidence to reject the null hypothesis

P-value

If $p \leq \alpha$: reject H_0

If $p > \alpha$: fail to reject H_0

1. One sample T-test

One sample t-test is used for comparison of the sample mean of the data to a particularly given value. We can use this when the sample size is small. (under 30) data is collected randomly and it is approximately normally distributed. It can be calculated as:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Diagram labels:

- Mean of the sample: $\bar{x}$
- Reference Value: μ
- Standard deviation: s
- Number of cases: $\sqrt{n}$

One sample t-test

Example: The weights of 25 obese people were taken before enrolling them into the nutrition camp. The population mean weight is found to be 45 kg before starting the camp. After finishing the camp for the same 25 people the sample mean was found to be 75 with a standard deviation of 25. Did the fitness camp work?

Python Implementation

Before its implementation we should have some basic knowledge about [numpy](#) and [scipy](#).

```
import scipy.stats as stats
import numpy as np

popu_mean = 45
s_mean = 75
s_std = 25
s_size = 25

t_stat = (s_mean - popu_mean) / (s_std / np.sqrt(s_size))
df = s_size - 1
alpha = 0.05
cr_t = stats.t.ppf(1 - alpha, df)

p_v = 1 - stats.t.cdf(t_stat, df)

print("T-Statistic:", t_stat)
print("Critical t-value:", cr_t)
print("P-Value:", p_v)

print('With T-value :')
if t_stat > cr_t:
    print("Significant difference. Camp had effect.")
else:
    print("No significant difference. Camp had no effect.")

print('With P-value :')
if p_v > alpha:
    print("Significant difference. Camp had effect.")
else:
    print("No significant difference. Camp had no effect.")
```

Output:

```
T-Statistic: 6.0
Critical t-value: 1.7108820799094275
P-Value: 1.703654035845048e-06
With T-value :
Significant difference. Camp had effect.
With P-value :
No significant difference. Camp had no effect.
```

Interpretation

- T-value (6.0) is much greater than the critical t-value (1.71), so we reject the null hypothesis.
- The [P-value](#) (0.0000017) is much smaller than 0.05, also indicating a significant result.

The fitness camp had a significant effect on participants weights, causing a measurable change.

2. Two sample T-test

2. Independent sample T-test

An Independent sample t-test commonly known as an unpaired sample t-test is used to find out if the differences found between two groups is actually significant or just a random occurrence. We can use this when:

- the population mean or standard deviation is unknown. (information about the population is unknown)
- the two samples are separate/independent. For i.e. boys and girls (the two are independent of each other)

It can be calculated using:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Diagram labels:

- Mean of the sample 1 (points to $\bar{X}_1$)
- Mean of the sample 2 (points to $\bar{X}_2$)
- Standard deviation sample 1 and 2 (points to S_1^2 and S_2^2)
- Number of cases sample 1 and 2 (points to n_1 and n_2)

Let's Take a example to understand

Researchers want to see if two teaching methods, A and B, produce different exam scores. Samples for both methods are collected independently.

Sample A (Teaching Method A): 78,84,92,88,75,80,85,90,87,79,78,84,92,88,75,80,85,90,87,79
Sample B (Teaching Method B): 82,88,75,90,78,85,88,77,92,80,82,88,75,90,78,85,88,77,92,80

Python Implementation

```
from scipy import stats
import numpy as np

A = np.array([78,84,92,88,75,80,85,90,87,79,78,84,92,88,75,80,85,90,87,79])
B = np.array([82,88,75,90,78,85,88,77,92,80,82,88,75,90,78,85,88,77,92,80])

t_val, p_val = stats.ttest_ind(A, B)

alpha = 0.05
df = len(A)+len(B)-2

crit_t = stats.t.ppf(1 - alpha/2, df)

print("T-value:", t_val)
print("P-Value:", p_val)
print("Critical t-value:", crit_t)

print('T-test Result:')
if np.abs(t_val) > crit_t:
    print('Significant difference found.')
else:
    print('No significant difference.')

print('P-test Result:')
if p_val > alpha:
    print('Fail to reject H0. No strong evidence of difference.')
else:
    print('Reject H0. Significant difference found.')
```

Output:

T-value: -0.008275847896130646
P-Value: 0.9934425963209128
Critical t-value: 2.0280940009804502
T-test Result:
No significant difference.
P-test Result:
Fail to reject H0. No strong evidence of difference.

Interpretation

- T-value (0.989) is less than the critical t-value (2.1009), so we fail to reject the null hypothesis.
- P-value (0.336) is greater than 0.05, meaning no significant difference.

There is no statistically significant difference between exam scores of Teaching Method A and Teaching Method B.

3. Paired Two-sample T-test

Paired sample t-test also known as dependent sample t-test is used to find out if the difference in the mean of two samples is 0. The test is done on dependent samples usually focusing on a particular group of people or things. In this each entity is measured twice resulting in a pair of observations.

We can use this when:

- Two similar samples are given. [i.e. Scores obtained in English and Math (both subjects)]
- The dependent variable data is continuous.
- The observations are independent of one another.
- The dependent variable is approximately normally distributed.

It can be calculated using

$$t = \frac{\bar{x}_d - 0}{\frac{s}{\sqrt{n}}}$$

Mean of the difference $\bar{x}_d$

Standard deviation s

Number of cases $\sqrt{n}$

paired t-test

Example Problem

Consider the following example. Scores (out of 25) of the subjects Math1 and Math2 are taken for a sample of 10 students. We have to perform the paired sample t-test for this data.

Math1: 4, 4, 7, 16, 20, 11, 13, 9, 11, 15
Math2: 15, 16, 14, 14, 22, 22, 23, 18, 18, 19

Python Implementation

```
from scipy import stats
import numpy as np

A = np.array([4, 4, 7, 16, 20, 11, 13, 9, 11, 15])
B = np.array([15, 16, 14, 14, 22, 22, 23, 18, 18, 19])

t_val, p_val = stats.ttest_rel(A, B)

alpha = 0.05
df = len(A)-1

c_t = stats.t.ppf(1 - alpha/2, df)

print("T-value:", t_val)
print("P-Value:", p_val)
print("Critical t-value:", c_t)

print('T-test:')
if np.abs(t_val) > c_t:
    print('Significant difference found.')
else:
    print('No significant difference.')

print('P-test:')
if p_val > alpha:
    print('Reject H0')
else:
    print('Fail to reject H0')
```

Output:

```
T-value: -4.953488372093023
P-Value: 0.0007875235561560145
Critical t-value: 2.2621571628540993
T-test:
Significant difference found.
P-test:
Fail to reject H0
```

Interpretation

- T-value (-4.95) is less than the negative critical t-value (-2.2622), so we reject the null hypothesis.
- P-value (0.00079) is less than 0.05, indicating significance.

There is a significant difference between Math1 and Math2 scores. The difference is unlikely due to chance.

